



# Shiming Wu

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## Education

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**The University of British Columbia**, Vancouver, Canada

PhD in Economics 2020 - 2026 (expected)

Master of Science in Business Administration in Finance 2018 - 2020

**Nankai University**, Tianjin, China

Bachelor of Economics in Finance 2012 - 2015

## Fields

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Insurance, Industrial Organization, Environmental Economics, Labor Economics, Machine Learning

## References

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**Paul Schrimpf**

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**Gorkem Bostanci**

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**David Green**

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**Prof. Gorkem Bostanci and Prof. Katherine Wagner want our department to submit letters for them. Please email [econ.grad@ubc.ca](mailto:econ.grad@ubc.ca) for reference letters. Thanks.**

## Research Grants, Awards, and Fellowships

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CIDER Small Grants in Innovative Data (C\$26,000) 2023 – 2024

Sir Quo-Wei Lee Fellowship from St. John's College 2024 - 2026

Harnan A.N. Singh Scholarship in Economics 2022

President's Academic Excellence Initiative PhD Award 2020 – 2026

Faculty of Arts Graduate Fellowship 2020 – 2025

International Student Tuition Award 2018 – 2026

Merit Student and Scholarships from Nankai University 2012 - 2015

## Job Market Paper

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### What Constrains Insolvency in Property Insurance?

Market Discipline, Capital Regulation, and Catastrophe Exposure

- This paper disentangles the roles of capital regulation and credit ratings in mitigating insolvency risk in the U.S. property insurance market. I first investigate the mechanism through which capital requirements affect the insurance market. Using reduced-form evidence and an instrumental variable approach that exploits a 2017 policy change as a quasi-experiment, I find that a \$1 million increase in required capital leads insurers to hold \$3.34 million more in capital and to raise insurance prices by 0.218 percentage points. These results reveal a direct trade-off between financial stability and consumer affordability. To further explore the underlying mechanisms, I develop a structural model in which insurers make capital and pricing decisions in a competitive market with limited liability and exposure to catastrophic risks.

Counterfactual analyses show that tightening capital requirements improves solvency but raises prices. In the absence of capital regulation, the model predicts that the insolvency rate would increase by 0.09 percentage points, while insurance prices would decline by about 5.1%, accompanied by greater risk-taking and market concentration. A third counterfactual scenario examines a market without capital regulation but with high credit rating salience. When consumers place greater emphasis on credit ratings, the insolvency rate decreases; however, intensified price competition reduces profitability and increases market concentration. Overall, the findings underscore that capital regulation remains crucial for sustaining market stability, as heightened rating salience alone cannot fully substitute for its stabilizing effects.

- In this paper, I apply several data science methods to structural estimation.
  - MLE to estimate return distribution and losses distributions
  - BLP type supply estimation to estimate random shocks of insolvency
  - Random forest to impute missing credit rating
  - Generalized additive model (GAM) to estimate capital and credit rating functions
  - I built a Linux workstation to run estimation

## Working Papers

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### Transformation of College to University and Impacts on Labor Market Outcomes

- This paper studies the impacts of different types of post-secondary education on education and labor market outcomes. I first use variations of distances to institutions and exogenous variations of colleges that upgraded into universities to investigate the value added to university education. Then I separately estimate the impacts of the transformed universities and traditional colleges and universities. I compared results from OLS and IV regressions. To address the treatment heterogeneity, I adapt the locally linear specification from Mountjoy (2022). Results suggest that the difference between university and college graduates is marginal. However, university entry improves labor market outcomes, such as employment and earnings compared with cohorts without post-secondary education. Graduates from transformed universities obtain a higher probability of being employed and higher earnings compared with people without post-secondary education. Nonetheless, transformed university graduates may have worse performance in the labor market, compared with graduates from colleges or traditional universities. Furthermore, graduates from transformed universities are more willing to register for training when they are unemployed compared with the other three groups.

### The Impacts of Health Insurance Mandates on the Labour Market and Health, Evidence from ACA

- This paper studies how employer mandates and health insurance affect labor market outcomes and health. I use staggered difference-in-differences research design and variations in the Affordable Care Act to learn how employer mandate affects labor market outcomes. I use doubly robust difference-in-differences in my main specifications to reduce selection bias. Results in the full sample suggest that the employer mandates in the Affordable Care Act increased hourly wages and did not have significant impacts on employment and part-time employment. Employer mandates stimulate a larger increase in employer-sponsored health insurance coverage rates among low-income workers. However, low-income workers are more vulnerable to involuntary part-time employment if employers reduce work hours to circumvent employer mandates. Firms prefer to reduce work hours to circumvent employer mandates instead of firing workers. Using doubly robust estimators and staggered difference-in-differences research design, I find evidence that providing health insurance improves workers' health. The employer mandate may increase productivity by improving workers' health status. Still, it may widen income inequality in the long run because low-income workers are more vulnerable to work hours losses.

## Work in Progress

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### **Mitigating the Trade-off: How Natural Catastrophe Adaptation Can Reconcile Insurer Solvency and Affordability**

- Stringent capital regulation in property insurance markets creates a tension between insurer solvency and affordability. This paper demonstrates that adaptation to natural catastrophes can mitigate this trade-off. I develop and estimate a model of the property insurance market with interdependent loss structures, competition, and capital regulation. My counterfactual analysis shows that adaptation measures, by reducing the correlation of losses, would decrease the market insolvency rate and lower market concentration. Furthermore, I find a net positive impact on social welfare when the costs of insurer insolvency are repurposed to subsidize adaptation. My results indicate that policy should focus on incentivizing adaptation as a key tool for maintaining sustainable insurance markets.

### **A Reinforcement Learning Approach to Dynamic Capital Regulation in Property Insurance**

- This paper examines the policy debate on transitioning from static, formula-based capital regulation in the U.S. property insurance market to a dynamic, model-based approach. The goal of this research is to discover an optimal, dynamic capital regulation policy that maximizes social welfare under profound uncertainty. Traditional counterfactual analysis relies on testing a few static policies against a handful of scenarios. To address this, I develop a learning-and-verification methodology that uses Reinforcement Learning to discover sophisticated, dynamic regulatory strategies, formal reasoning to verify compliance with safety constraints, and decision-tree extraction to produce interpretable policy rules. This hybrid approach moves beyond the limitations of manual analysis by autonomously exploring a vast policy space over millions of simulated futures, yielding strategies that are inherently far-sighted and robust. The results suggest that such a model-based regime can enhance the resilience and adaptability of the insurance sector while maintaining regulatory accountability and transparency.

### **From Monoculture to Market: The Economics of Plant Biodiversity and Consumer Welfare**

- This paper investigates the complex relationship between industrial agriculture, plant biodiversity, and economic welfare. While industrial agriculture has been credited with increasing food production, it has also been linked to a significant decline in agrobiodiversity. This research disentangles two opposing economic forces: the homogenizing effect of industrial agriculture's focus on a few high-yield, storable crops, and the potential for market competition to foster product variety. I develop and estimate a structural model of demand and supply for fresh produce, employing a random coefficients logit model (BLP) to capture heterogeneous consumer preferences for a wide range of crop attributes, including variety. On the supply side, I model farmers' and food companies' decisions to offer different cultivars, considering the influence of production costs, market structure, and agricultural policies. Using detailed market-level data, I quantify the welfare effects of changes in plant biodiversity on consumers and producers. Furthermore, I extend the traditional welfare analysis to incorporate the non-market value of agrobiodiversity as a source of resilience to climate shocks. My findings aim to provide a more complete picture of the true social costs and benefits of industrial agriculture and to inform policies that promote a more diverse and sustainable food system.

## Teaching

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ECON 502 Macroeconomics, teaching TA (UBC)

ECON 490 Seminar in Applied Economics, TA (UBC)

ECON 370 Benefit-Cost Analysis and the Economics of Project Evaluation, TA (UBC)

ECON 356 Introduction to International Finance, TA (UBC)

ECON 255 Understanding Globalization, teaching TA (UBC)

ECON 102 Principles of Macroeconomics, teaching TA (UBC)

ECON 101 Principles of Microeconomics, teaching TA (UBC)

Conferences

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2024: 19th CIREQ PhD Students’ Conference  
2022: NBER Insurance Working Group Meeting

Employment

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|--------------------------------------------------------------|-----------------------------------|
| Teaching assistant at University of British Columbia         | 2021 - present, Vancouver, Canada |
| Research assistant at University of British Columbia         | 2019 - 2024, Vancouver, Canada    |
| Tutor and teaching assistant at Yingde Inspiration Education | 2017 - 2018, Yingde, China        |
| Auditor at KPMG                                              | 2015 - 2016, Beijing, China       |

Other Information

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|-----------------|----------------------------------------------------------------------------|
| Computer Skills | R, Julia, Stata, Matlab, SPSS, Python, L <sup>A</sup> T <sub>E</sub> X     |
| Data Science    | Machine learning, Causal inference, Structural estimation                  |
| Languages       | English (fluent), Mandarin (native), Cantonese (native), French (learning) |
| Certificates    | Passed CFA Level II Exam                                                   |
| Datasets        | Bloomberg, Capital IQ                                                      |
| Gender          | Female                                                                     |